Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A buffer solution has equal volume of $0.2 \mathrm{MNH_4OH}$ and $0.02 \mathrm{MNH_4Cl}$. The $\mathrm{pK_b}$ of the base is 5. The pH is

(A) 4

(B) 10

(C) 7

(D) 9

Q.2

The following equilibrium is established when hydrogen chloride is dissolved in acetic acid.

$\mathrm{HCl + CH_3COOH = Cl^- + CH_3COOH_2^+}$

The set that characterises the conjugate acidbase pairs is

(A) $\left(\mathrm{HCl, CH_3COOH}\right)$ and $\left(\mathrm{CH_3COOH_2^+, Cl^-}\right)$

(B) $\left(\mathrm{HCl, CH_3COOH_2^+}\right)$ and $\left(\mathrm{CH_3COOH, Cl^-}\right)$

(C) $\left(\mathrm{CH_3COOH_2^+, HCl}\right)$ and $\left(\mathrm{Cl^-, CH_3COOH}\right)$

(D) $\left(\mathrm{HCl, Cl^-}\right)$ and $\left(\mathrm{CH_3COOH_2^+, CH_3COOH}\right)$

Q.3

Assuming that the buffer in the blood $1 \mathrm{~s}$ $\mathrm{CO_2 - HCO_3^-}$. Calculate the ratio of conjugate base to acid necessary to maintain blood at its proper pH of 7.4 $\mathrm{K_1\left(H_2CO_3\right) = 4.5 \times 10^{-7}}$

(A) 11

(B) 8

(C) 6

(D) 14

Q.4

In a buffer solution having equal concentrations of B^- and HB the value of K_b for B^- is 10^{-9}. The pH of buffer solution is

(A) 9

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(B) 5

(C) 4

(D) 6

Q.5

For preparing a buffer solution of $\mathrm{pH6}$ by mixing sodium acetate and acetic acid, the ratio of the concentration of salt and acid should be $\left(\mathrm{K_a} = 10^{-5}\right)$

(A) $1:10$

(B) $10:1$

(C) $100:1$

(D) $1:100$

Q.6

Calculate the number of moles of NH_4Cl that should be added to one litre of 1.0 M NH_4OH to prepare buffer solution with $pH=9 K_b=2\times10^{-5}$,takelog2=0.3

(A) 3.4

(B) 2.6

(C) 1.5

(D) 2

Q.7

What will be the H^+ ion concentration in a solution prepared by mixing $50\ \mathrm{mL}$ of $0.20\mathrm{MNaCl}$, $25\ \mathrm{mL}$ of $0.10\mathrm{MNaOH}$ and $25\ \mathrm{mL}$ of $0.30\mathrm{NHCl}$?

(A) $0.5\mathrm{M}$

(B) $0.05\mathrm{M}$

(C) $0.02\mathrm{M}$

(D) $0.10\mathrm{M}$

Q.8

The dissociation constant of a weak acid is 1×10^{-4}. In order to prepare a buffer solution with a $\mathrm{pH} = 5$ the [Salt] $/$ [Acid] ratio should be

(A) $1:10$

(B) $4:5$

(C) $10:1$

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(D) $5:4$

Q.9

Calculate the pH of $0.5\mathrm{M}$ aqueous solution of NaCN, the $\mathrm{pK_b}$ of $\mathrm{CN^-}$ is 4.70

(A) 12.5

(B) 13.5

(C) 10.5

(D) 11.5

Q.10

Which of the following salt does not get hydrolysed in water

(A) $\mathrm{KClO_4}$

(B) $\mathrm{NH_4Cl}$

(C) $\mathrm{CH_3COONa}$

(D) None of these

Q.11

Equimolar solutions of the following compounds are prepared separately in water. Which will have the lowest pH value ?

(A) $\mathrm{BeCl_2}$

(B) $\mathrm{SrCl_2}$

(C) $\mathrm{CaCl_2}$

(D) $\mathrm{MgCl_2}$

Q.12

What is the percentage hydrolysis of NaCN in $\mathrm{N}/80$ solution when the dissociation constant for HCN is 1.3×10^{-9} and $\mathrm{K_w} = 1.0 \times 10^{-14}$

(A) 2.48

(B) 5.26

(C) 8.2

(D) 9.6

Q.13

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Given: K_b (dimethylamine) $= 7.4 \times 10^{-4}\mathrm{M}$. A buffer is made by mixing $(CH_3)_2NH$ and $(CH_3)_2NH_2Cl$. The range of buffer to which this buffer can be used is

(A) 2.13 to 4.13

(B) 6.13 to 8.13

(C) 8.13 to 10.13

(D) 9.87 to 11.87

Q.14

If $K_{a1}\left(H_2CO_3\right) = 4.5 \times 10^{-7}\mathrm{M}$ and $K_a\left(HCO_3^-\right) = 5.0 \times 10^{-11}\mathrm{M}$, the pH of $0.1\mathrm{M}Na_2CO_3$ solution at $25^\circ C$ will be

(A) 8.2

(B) 9.4

(C) 10.5

(D) 11.7

Q.15

The pH of $0.1\mathrm{M}NaHCO_3$ solution can be computed from the expression

(A) $\mathrm{pH} = \mathrm{p}K_{a1}^\circ + K_{a2}^\circ$

(B) $\mathrm{pH} = \frac{1}{2}\left(\mathrm{p}K_{a1}^\circ + \mathrm{p}K_{a2}^\circ\right)$

(C) $\mathrm{pH} = \mathrm{p}K_{a1}^\circ - K_{a2}^\circ$

(D) $\mathrm{pH} = \frac{1}{2}\left(\mathrm{p}K_{a1}^\circ - K_{a2}^\circ\right)$

Answers & Solutions

Q.1 Answer:

10

Solution:

$$\mathrm{pOH} = \mathrm{pK_b} + \log_{10}\frac{[\mathrm{Salt}]}{[\,\mathrm{Base}\,]}$$
$$= 5 + \log_{10}\frac{0.02}{0.2} = 5 + (-1) = 4$$
$$\mathrm{pH} = 14 - \mathrm{pOH} = 14 - 4 = 10$$

Q.2 Answer:

$\left(HCl, Cl^-\right)$ and $\left(CH_3COOH_2^+, CH_3COOH\right)$

Solution:

HCl is stronger acid than CH_3COOH and Cl^- is a stronger base than $CH_3COOH_2^+$ and is the conjugate base of HCl.

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$$\underset{\text{acid}_1}{HCl} + \underset{\text{base}_2}{CH_3COOH} \rightleftharpoons \underset{\text{base}_1}{Cl^-} + \underset{\text{acid}_2}{CH_3COOH_2^+}$$

Q.3 Answer:

11

Solution:

CO_2 **with** H_2O **forms** H_2CO_3

$$CO_2 + H_2O \rightleftharpoons H^+ + HCO_3^-$$

$$K_1 = \frac{[H^+][HCO_3^-]}{[CO_2]} = 4.5 \times 10^{-7}$$

Again $pH = -\log[H^+] = 7.4$

$$\therefore \quad [H^+] = 4.0 \times 10^{-8}$$

$$\therefore \quad \frac{[HCO_3^-]}{[CO_2]} = \frac{4.5 \times 10^{-7}}{4 \times 10^{-8}} = 11$$

Q.4 Answer:

5

Solution:

For acidic buffer pH can be find out as follows

$$pH = pKa + \log_{10}\frac{[\text{ salt }]}{[\text{ acid }]}$$

As here $K_b = 10^{-9}$

So $K_a = 10^{-5}$

$pK_a = 5$

As $pH = 5 + \log_{10}\frac{[B^-]}{[HB]}$

As $[B^-] = [HB]$

$pH = 5 + \log 1 = 5 + 0 = 5$

Q.5 Answer:

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$10:1$

Solution:

$6 = -\log 10^{-5} + \log \frac{\text{salt}}{\text{acid}} = 5 + \log \frac{\text{salt}}{\text{acid}}$

$\log \frac{\text{salt}}{\text{acid}}$ **must be 1.** $\therefore \frac{\text{salt}}{\text{acid}} = \frac{10}{1}$ **or** $10:1$

Q.6 Answer:

2

Solution:

It is basic buffer

$pOH = pK_b + \log \frac{[salt]}{[weakbase]}$

$pOH = 14 - 9 = 5$

$5 = 4.7 + \log \frac{x}{1}$

$0.3 = \log x \Rightarrow x = 2$

Q.7 Answer:

$0.05M$

Solution:

$[H^+] = \frac{(25\times0.3-25\times0.1)}{100} meq = 0.05M$

Q.8 Answer:

$10:1$

Solution:

$pH = pK_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$

$5 = 4 + \log \frac{[Salt]}{[Acid]} \quad [\because pK_a = -\log K_a]$

Given, $K_a = 1 \times 10^{-4}$

$\therefore pK_a = -\log\left(1 \times 10^{-4}\right) = 4$

Now from Handerson equation $pH = pK_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$

Putting the values $5 = 4 + \log \frac{[\text{Salt}]}{[\text{Acid}]}$

$\log \frac{[Salt]}{[\text{Acid}]} = 5 - 4 = 1$

Taking antilog $[\text{Salt}]/[\text{Acid}] = 10 = 10:1$

Q.9 Answer:

11.5

Solution:

$NaCN$ **is a salt of strong base and weak acid ;** $pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$

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pK_a for $HCN = 14 - 4.70 = 9.30$

$\therefore pH = 7 + \frac{1}{2} \times 9.30 + \frac{1}{2} \log 0.5; pH = 11.5$

Q.10 Answer:

$KClO_4$

Solution:

$$KClO_4 \rightleftharpoons K^+ + ClO_4^-; \; H_2O \rightleftharpoons \underset{\substack{\Downarrow \\ KOH \\ \text{Strong}}}{OH^-} + \underset{\substack{\Downarrow \\ HClO_4 \\ \text{Strong}}}{H^+}$$

Q.11 Answer:

$BeCl_2$

Solution:

Metal halide on hydrolysis with water form corresponding hydroxides. The basic strength of hydroxide increases as we move down in a group. This is because of the increase in size which results in decrease of ionization energy which weakens the strength of $M - O$ bonds in MOH and thus increases the basic strength.

$\underset{\text{Amphoteric}}{Be(OH)_2}$ $\underset{\text{Weak base}}{Mg(OH)_2}$

$\underset{\text{Strong base}}{Ca(OH)_2 \; Sr(OH)_2 \; Ba(OH)_2}$

Hence, $Be(OH)_2$ will have lowest pH.

Q.12 Answer:

2.48

Solution:

$$\alpha = \sqrt{\frac{K_w}{K_a \times c}} = \sqrt{\frac{10^{-14}}{1.3 \times 10^{-9} \times \frac{1}{80}}} = 2.48\%.$$

Q.13 Answer:

9.87 to 11.87

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Solution:

$\mathrm{p}K_{\mathrm{b}}^{\circ} = -\log\left(7.4 \times 10^{-4}\mathrm{M}\right) = 3.13$

Buffer range is $14 - \mathrm{p}K_{\mathrm{b}} - 1$ **to** $14 - \mathrm{p}K_{\mathrm{b}} + 1$**, i.e.** 9.87 **to** 11.87

Q.14 Answer:

11.7

Solution:

$\mathrm{CO}_3^{2-} + \mathrm{H_2O} \rightleftharpoons \mathrm{HCO}_3^- + \mathrm{OH}^-;$

$\mathrm{pH} = \frac{1}{2}\left[\mathrm{p}K_{\mathrm{w}}^{\circ} + \mathrm{p}K_{\mathrm{a}}^{\circ}\left(\mathrm{HCO}_3^-\right) + \log(c/\mathrm{M})\right]$

$= \frac{1}{2}\left[14 - \log\left(5 \times 10^{-11}\right) + \log(0.1)\right] = \frac{1}{2}[14 + 10.30 - 1] = 11.65$

Q.15 Answer:

$\mathrm{pH} = \frac{1}{2}\left(\mathrm{p}K_{\mathrm{a1}}^{\circ} + \mathrm{p}K_{\mathrm{a2}}^{\circ}\right)$

Solution:

Na^+ **does not show hydrolysis reaction.** HCO_3^- **is an amphiprotic anion.**

$\mathrm{HCO}_3^- + \mathrm{H_2O} \rightleftharpoons \mathrm{H_3O}^+ + \mathrm{CO}_3^{2-}; \quad K_{\mathrm{a2}} = \frac{\left[\mathrm{H_3O}^+\right]\left[\mathrm{CO}_3^{2-}\right]}{\left[\mathrm{HCO}_3^-\right]}$

$\mathrm{HCO}_3^- + \mathrm{H_2O} \rightleftharpoons \mathrm{H_2CO_3} + \mathrm{OH}^-; \quad K_{\mathrm{b}} = \frac{\left[\mathrm{H_2CO_3}\right]\left[\mathrm{OH}^-\right]}{\left[\mathrm{HCO}_3^-\right]}$

Also $K_{\mathrm{b}} = \frac{\left[\mathrm{OH}^-\right]\left[\mathrm{H}^+\right]}{\left[\mathrm{HCO}_3^-\right][\mathrm{H}^+]/[\mathrm{H_2CO_3}]} = \frac{K_{\mathrm{w}}}{K_{\mathrm{a1}}}$

From the expression of K_{a2}**, we get**

$\left[\mathrm{H_3O}^+\right] = \frac{K_{\mathrm{a2}}\left[\mathrm{HCO}_3^-\right]}{\left[\mathrm{CO}_3^{2-}\right]} = \frac{K_{\mathrm{a2}}\left\{\left[\mathrm{H_2CO_3}^-\right]\left[\mathrm{OH}^-\right]/K_{\mathrm{b}}\right\}}{\left[\mathrm{CO}_3^{2-}\right]} = \frac{K_{\mathrm{a2}}}{\left[\mathrm{CO}_3^{2-}\right]}\frac{[\mathrm{H_2CO_3}]}{(K_{\mathrm{w}}/K_{\mathrm{a1}})}\frac{K_{\mathrm{w}}}{[\mathrm{H_3O}^+]}$

or $\left[\mathrm{H_3O}^+\right]^2 = K_{\mathrm{a1}}K_{\mathrm{a2}}\frac{[\mathrm{H_2CO_3}]}{\left[\mathrm{CO}_3^{2-}\right]}$

Since both K_{a2} **and** K_{b} **are usually very small, one may assume** $\left[\mathrm{H_2CO_3}\right] \simeq \left[\mathrm{CO}_3^{2-}\right]$**. Hence**

or $\left[\mathrm{H_3O}^+\right]^2 = K_{\mathrm{a1}}K_{\mathrm{a2}}$ **or** $\mathrm{pH} = \frac{1}{2}\left(\mathrm{p}K_{\mathrm{a1}}^{\circ} + \mathrm{p}K_{\mathrm{a2}}^{\circ}\right)$

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